Aiming for Competitive Balance

Designing Fair Handicap Systems for Darts using a Markov Decision Process Framework

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Joint work with

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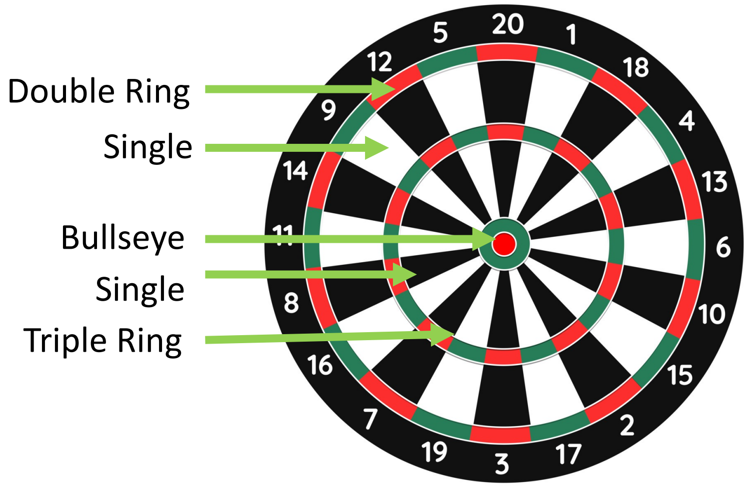
The Rules of Darts

- Two players with starting score of 501
- Race to zero
- To “check out”, must hit a “double” and result in score of exactly zero
- Can also “bust” by landing on a score where checking out is impossible



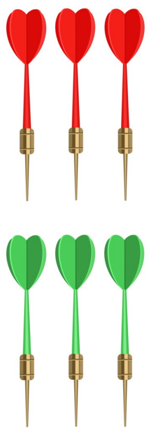
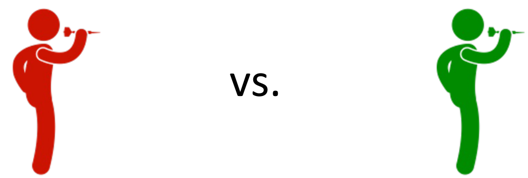


Image Sources: https://www.vecteezy.com/free-vector/dart-game and https://thenounproject.com/icon/darts-642047/

Competitive Imbalance and Handicap Systems

- Wide range of skill levels



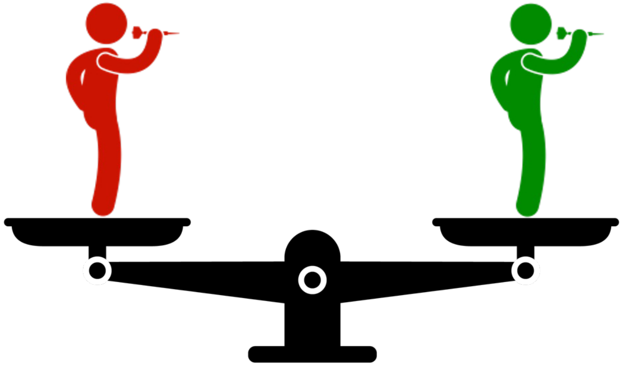
- **Handicap systems** create competitive balance between mismatched players

- Existing darts handicaps use heuristics to give weaker player a head-start

Image Sources: https://www.flaticon.com/free-icon/start-line_2078790 and https://www.flaticon.com/free-icon/finish-line-flag_1647

Research Goals

- Develop optimization-driven handicapping for darts
- Build intuition around fairness in darts
- Model a novel dynamic credit handicap in darts (Chan and Singal, 2016)
  - Credit: deterministically choose outcome for single "throw"
- Propose a handicap system that mathematically balances competition



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Modelling Framework

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Model: Markov Decision Process (MDP)

Player

State

$s \in S$

Environment

Model: Markov Decision Process (MDP)

State

$s \in S$

1. Player's Remaining Score

$\{0, \dots, 501\}$

2. Player's Remaining Credits

$\{0, \dots, 9\}$

Model: Markov Decision Process (MDP)

Player

State
$s \in S$

Action $a \in A$

Environment

Model: Markov Decision Process (MDP)

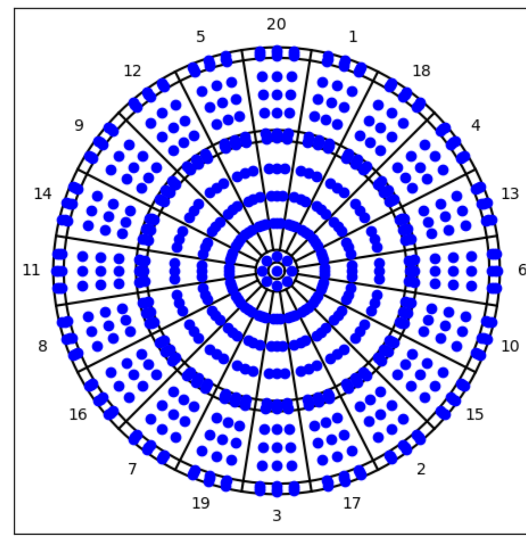
Action

$a \in A$

$$A = \{A_{throw} \cup A_{credit}\}$$

A_{throw} contains 729 targets on the board that a player may aim for

A_{credit} contains the 62 possible board regions that a player may "hit" with perfect execution



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Model: Markov Decision Process (MDP)

Player

State
$s \in S$

Action $a \in A$

Environment

Transition
Probabilities
$p(s'|s, a)$

Model: Markov Decision Process (MDP)

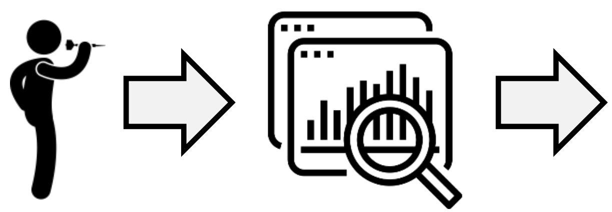
Transition Probabilities

$$p(s'|s,a)$$

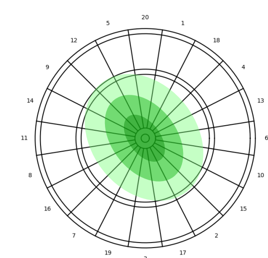
$a \in A_{throw}$

Uncertain Outcome

Use professional player's skill model*



$\mathcal{N}(\mu, \Sigma)$



$a \in A_{credit}$

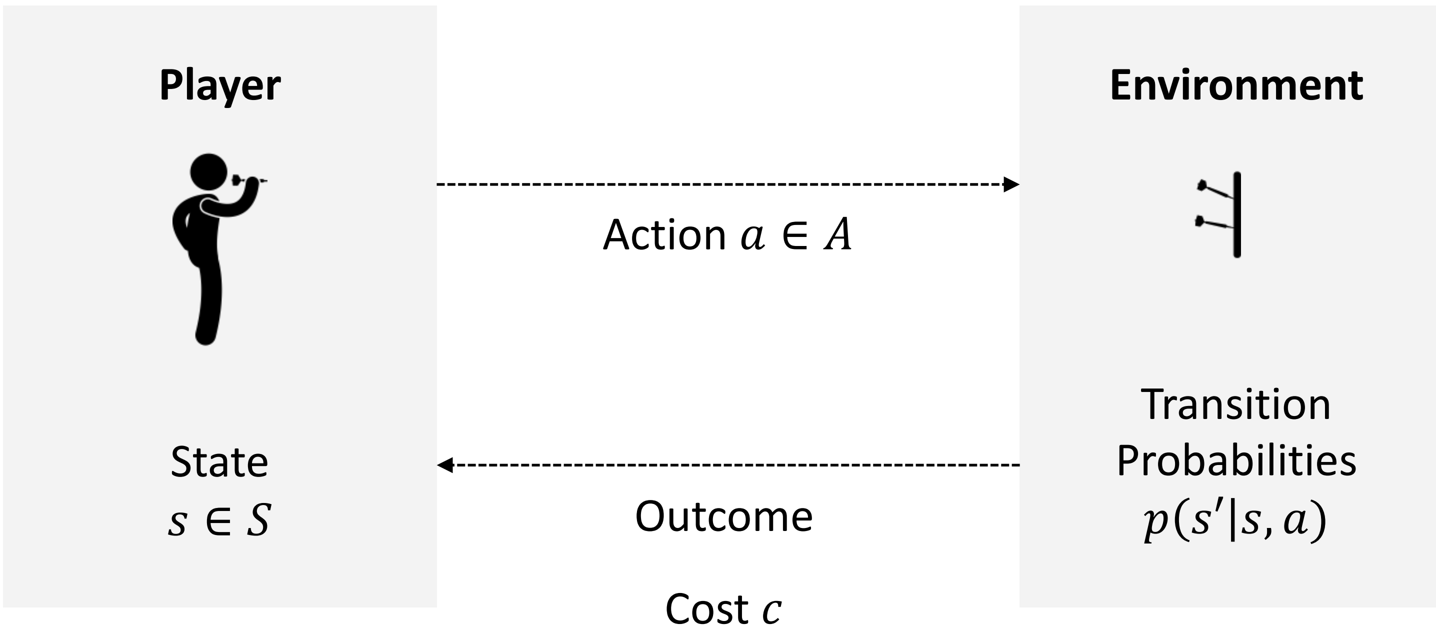
Deterministic Outcome

Hit desired board region with certainty

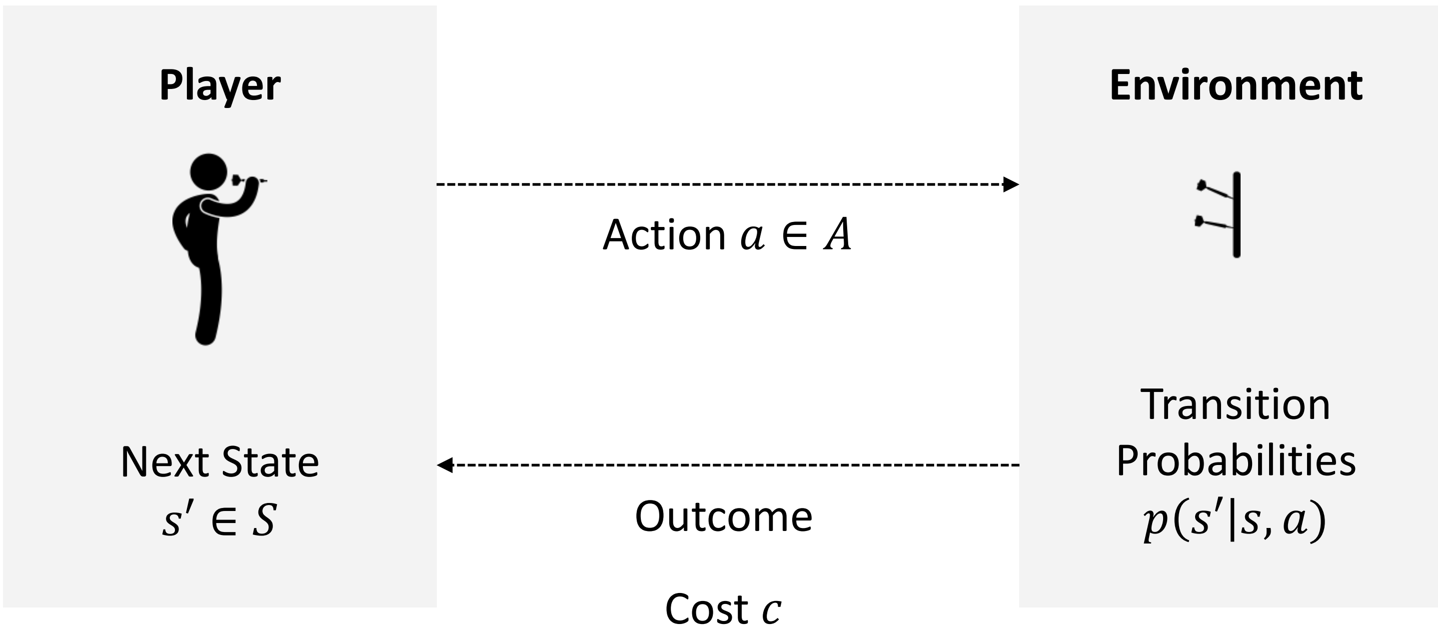
*(Haugh and Wang, 2022)

Image Sources: https://www.flaticon.com/free-icon/data_993762

Model: Markov Decision Process (MDP)



Model: Markov Decision Process (MDP)



Model: Different Skill Levels

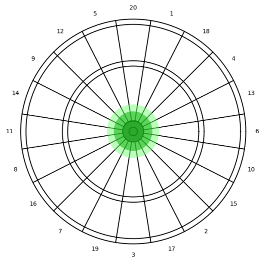
Different skill levels are modelled using an **execution error multiplier ϵ** (Chan et al, 2022)

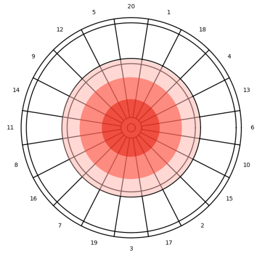
$\mathcal{N}(\mu, \Sigma)$

$\mathcal{N}_{\epsilon}(\mu, \epsilon\, \Sigma)$

Strongest Player
$\boldsymbol{\epsilon = 1}$

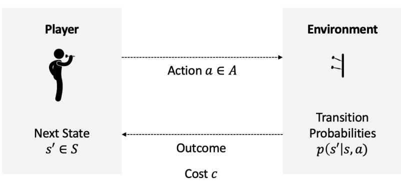
Weaker Players
$\boldsymbol{\epsilon > 1}$





Model: Markov Decision Process (MDP)

Model Definition



Bellman Equation

$$V_\epsilon^\pi(s) = c(s,\pi(s)) + \sum_{s' \in S} P_\epsilon(s'|s,\pi(s)) V_\epsilon^\pi(s')$$

Solution

$\pi_\epsilon^*(s)$ Optimal Policy

$V_\epsilon^*(s)$ Optimal Value

Value Function Interpretation

Theorem 1
Value function is the expected number of throws for a player to check out when using a given policy.

Definition 1
The **weaker player** has a higher expected throws to check out when starting from a score of 501 (with no credits).

Definition 2
Competitive balance occurs when two players have an equal number of expected throws to check out.

Theorem 2
There exists a unique number of credits that gives a weaker player the needed advantage to check out with equal or less expected throws compared to a stronger player

Results*

* Solved a more complex model formulation which includes the turn feature of the game (i.e., players alternate every 3 throws).

Solving without Dynamic Credits

$\mathcal{N}_1(\mu, \Sigma)$	$\mathcal{N}_2(\mu, 2\Sigma)$		$\mathcal{N}_8(\mu, 8\Sigma)$	$\mathcal{N}_9(\mu, 9\Sigma)$
$\boldsymbol{\epsilon = 1}$	$\boldsymbol{\epsilon = 2}$	…	$\boldsymbol{\epsilon = 8}$	$\boldsymbol{\epsilon = 9}$

No Credits

Strongest

Weakest

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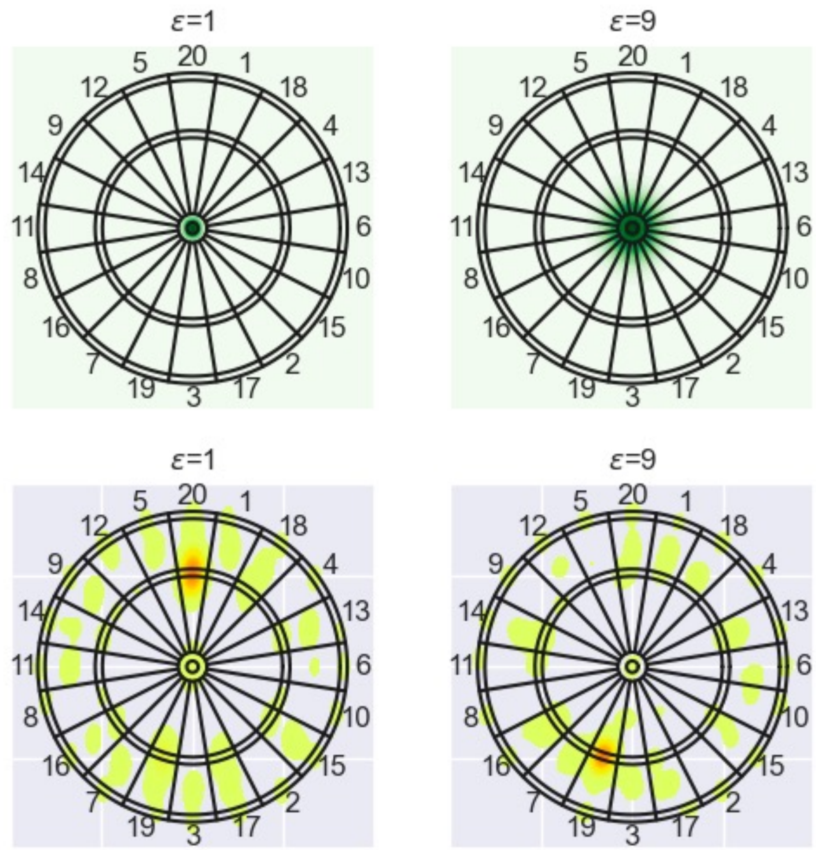
Results: Execution Error and Optimal Policy

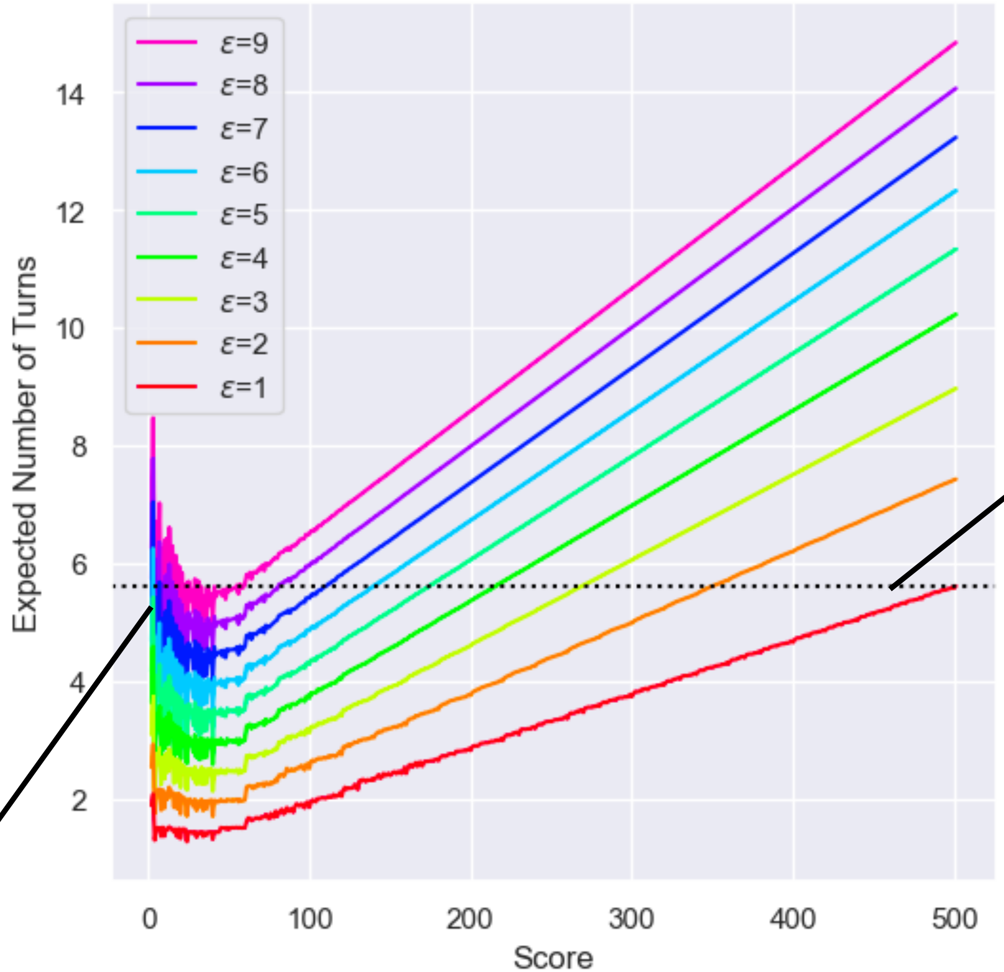
Skill Model
(target = bullseye)

Optimal Policy



Distribution of throw outcomes (top) vs. optimal policy (bottom) for players with $\epsilon = 1$ (left) and $\epsilon = 9$ (right)

Results: Sources of Imbalance

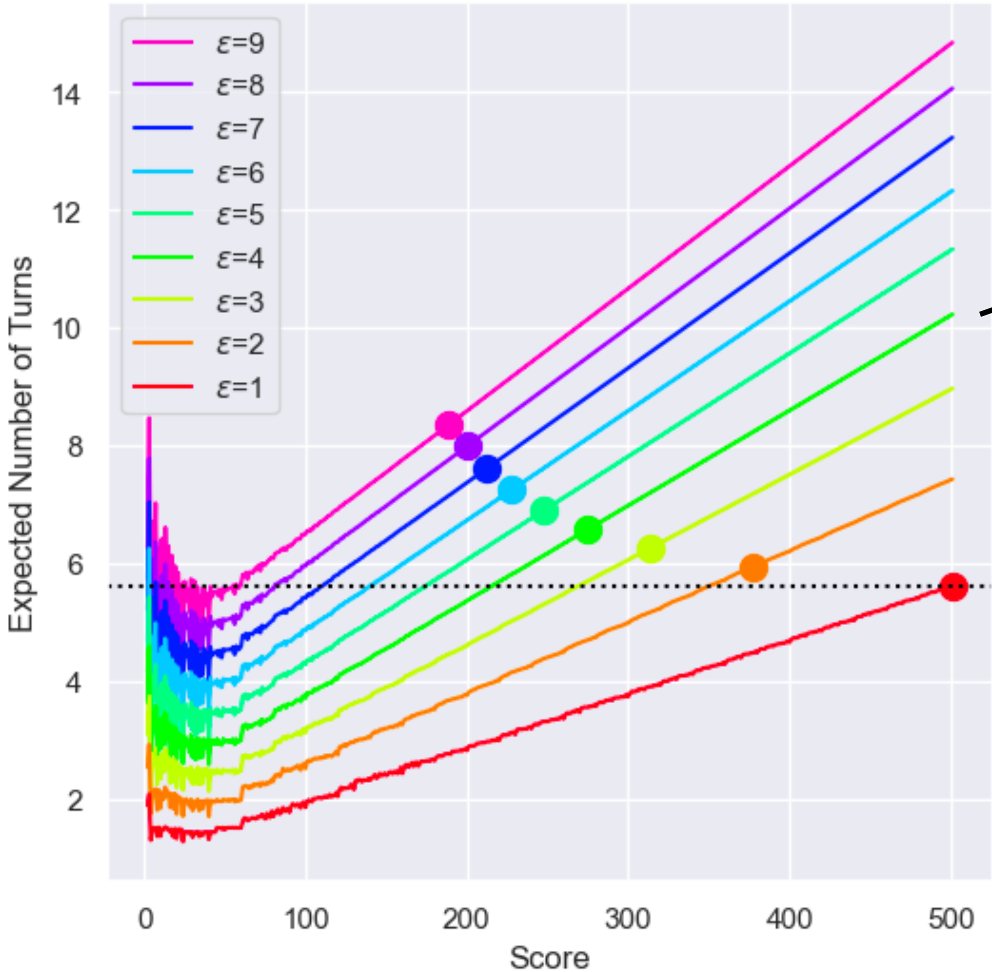


Main imbalance at end of game

Stronger player could dominate across all scores (head-start could never balance competition)

Expected turns remaining at different scores for players with different execution error

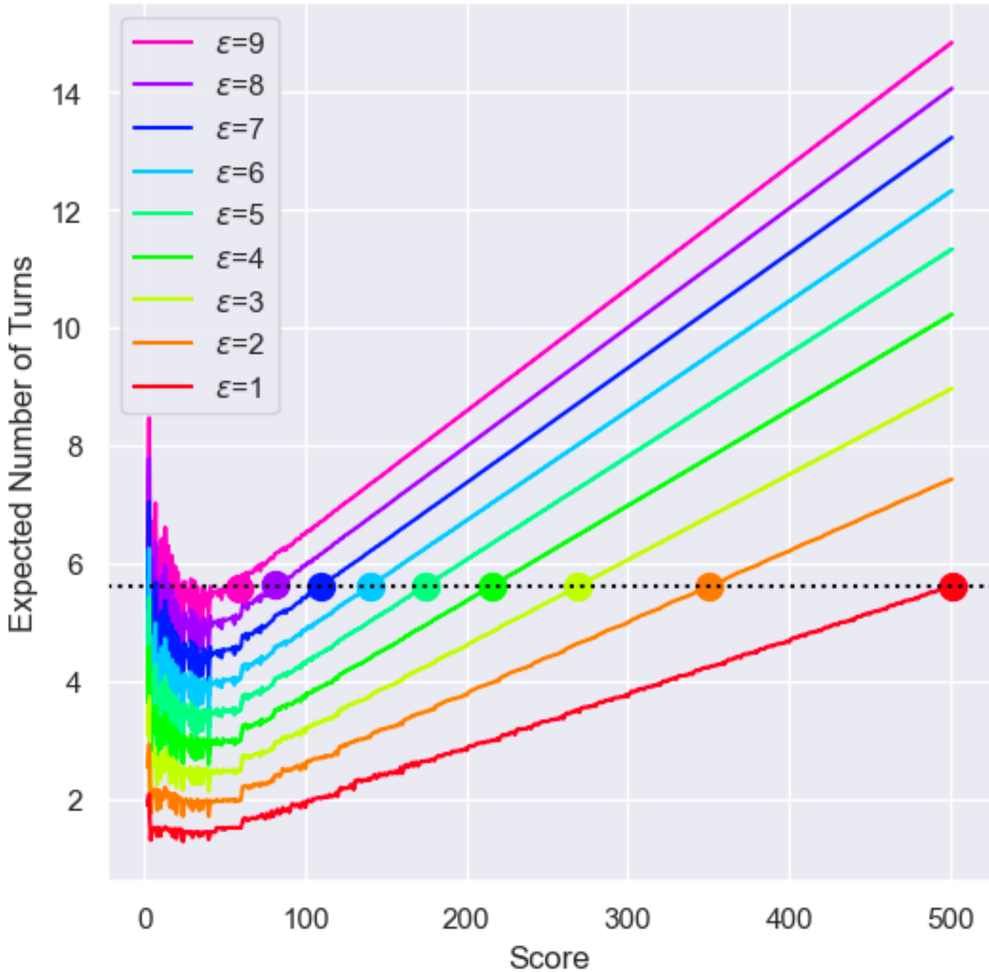
Results: Heuristic Head-Start Handicap



Current heuristic head-start handicaps do not balance competition

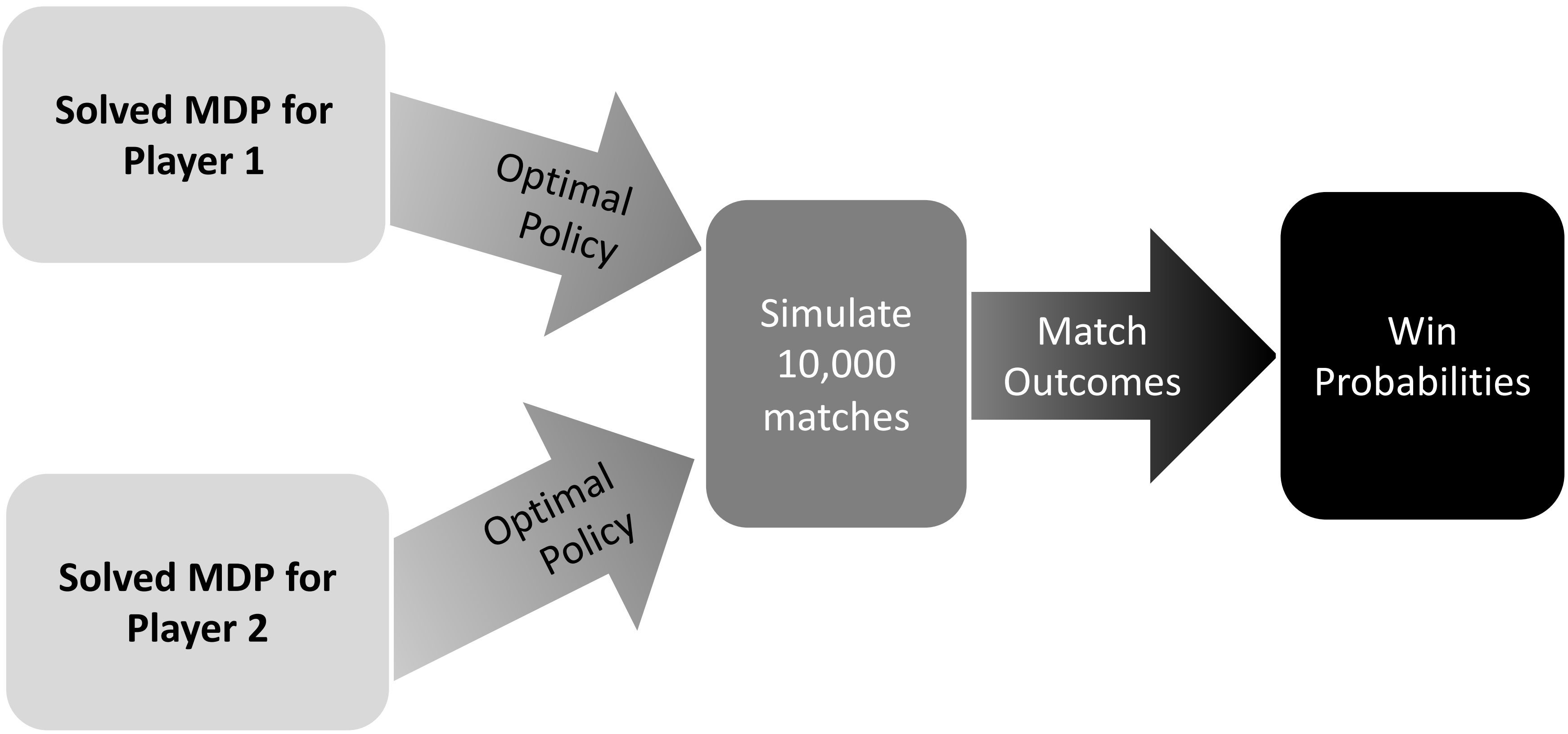
Heuristic spot point handicap when stronger player has $\epsilon = 1$ (markers)

Results: Optimized Head-Start Handicap

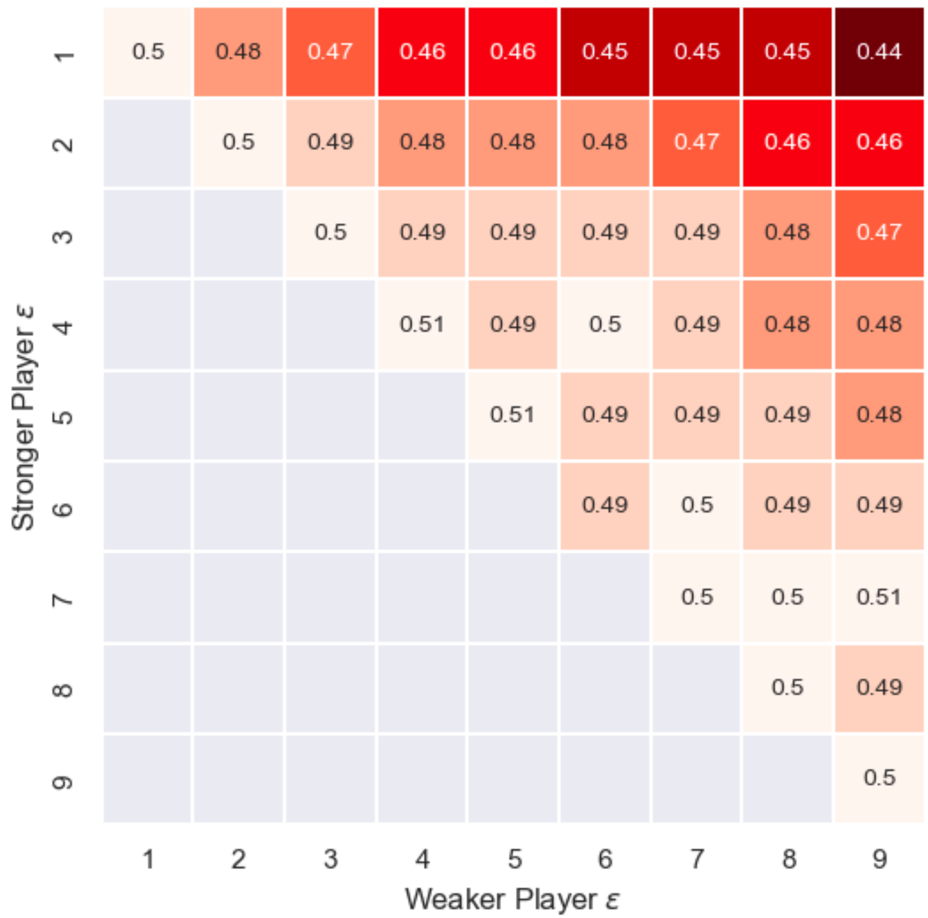


Optimized spot point handicap when stronger player has $\epsilon = 1$ (markers)

Testing Handicaps with Simulation



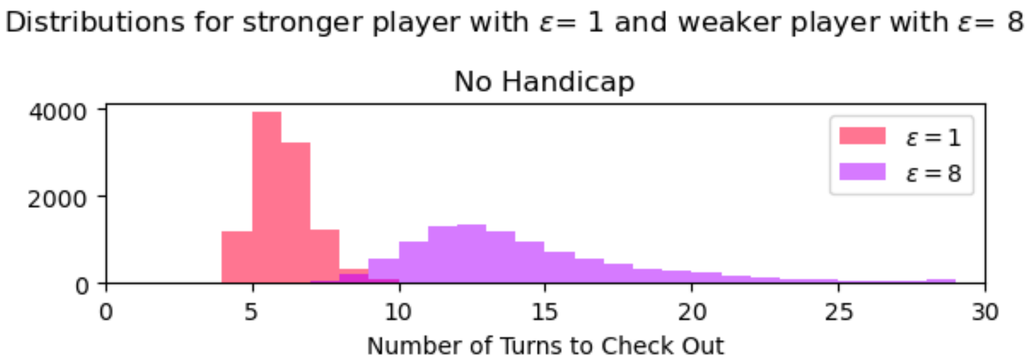
Results: Optimized Head-Start Handicap



Stronger player wins less than 50% of the time using this handicap

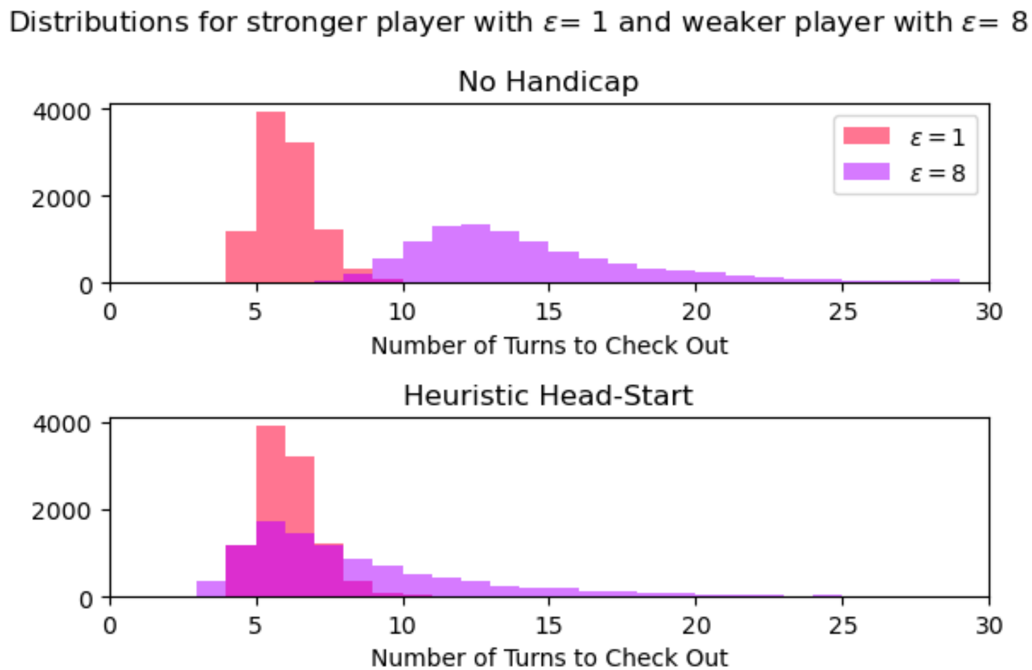
Probability that the stronger player wins after 10,000 simulated matches for different skill mismatches

Results: Optimized Head-Start Handicap



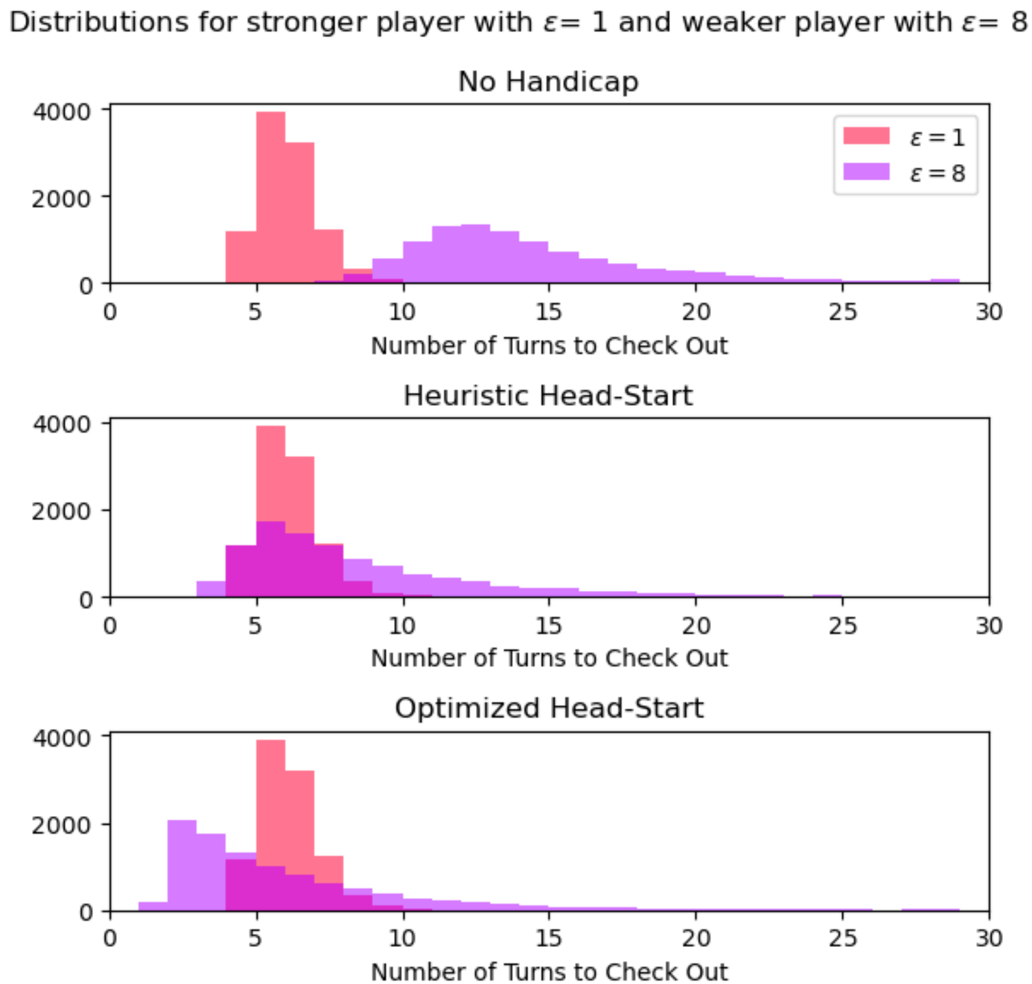
Number of turns for two mismatched players in simulation with 10,000 iterations

Results: Optimized Head-Start Handicap



Number of turns for two mismatched players in simulation with 10,000 iterations

Results: Optimized Head-Start Handicap



Number of turns for two mismatched players in simulation with 10,000 iterations

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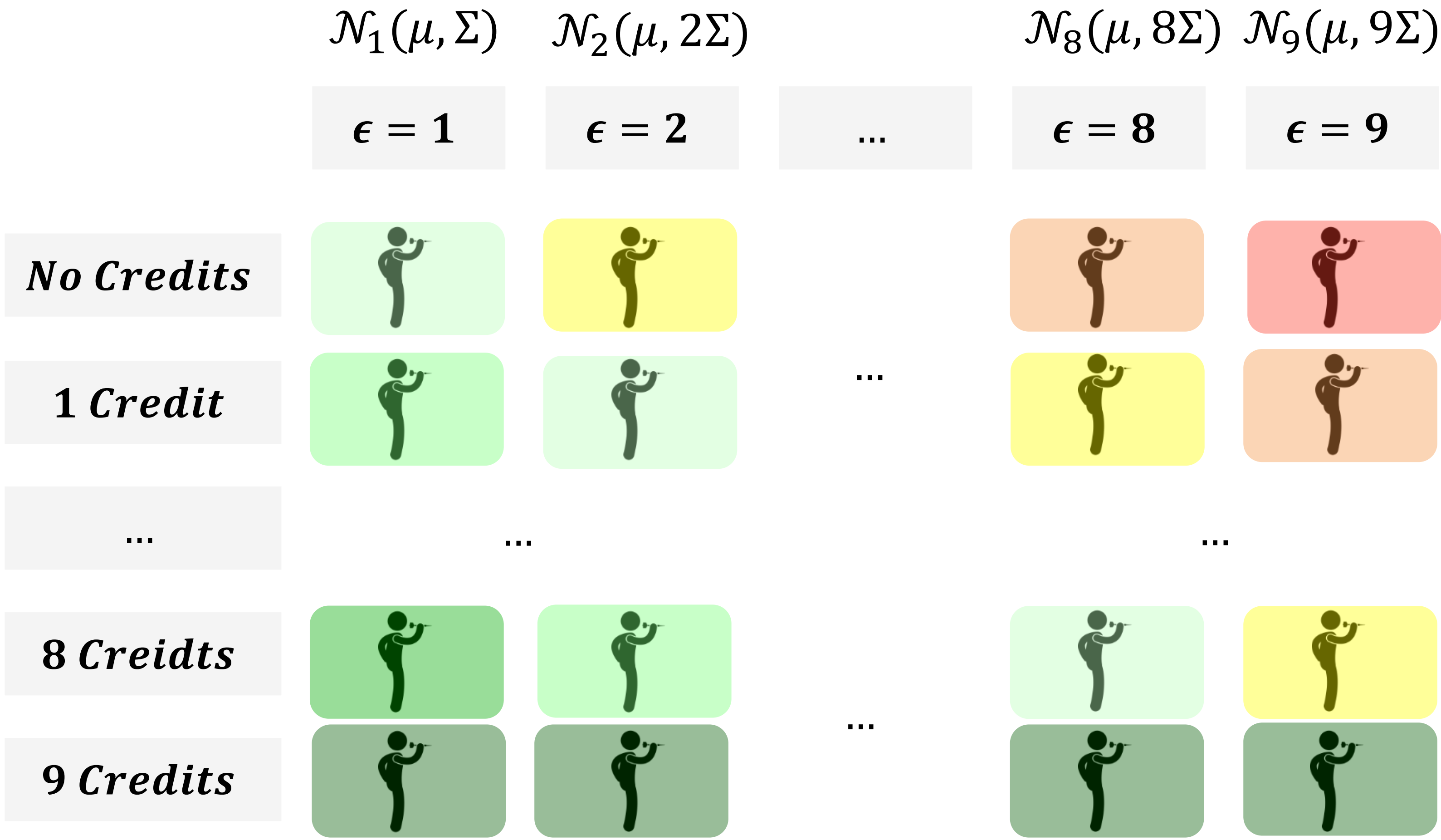
Adding Dynamic Credits



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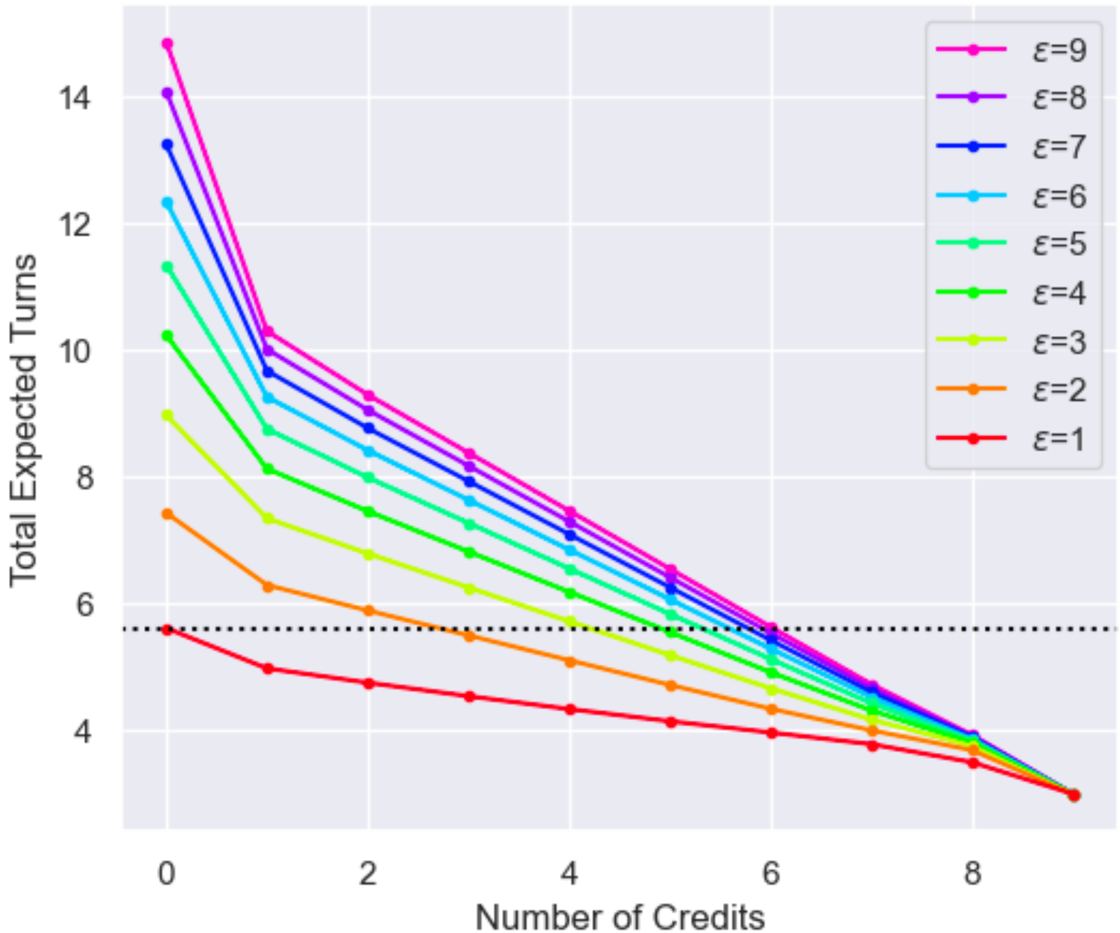
Results: Dynamic Credits



Dynamic credit handicap finds the number of credits where the weaker player and stronger player have the same expected number of turns starting from a score of 501

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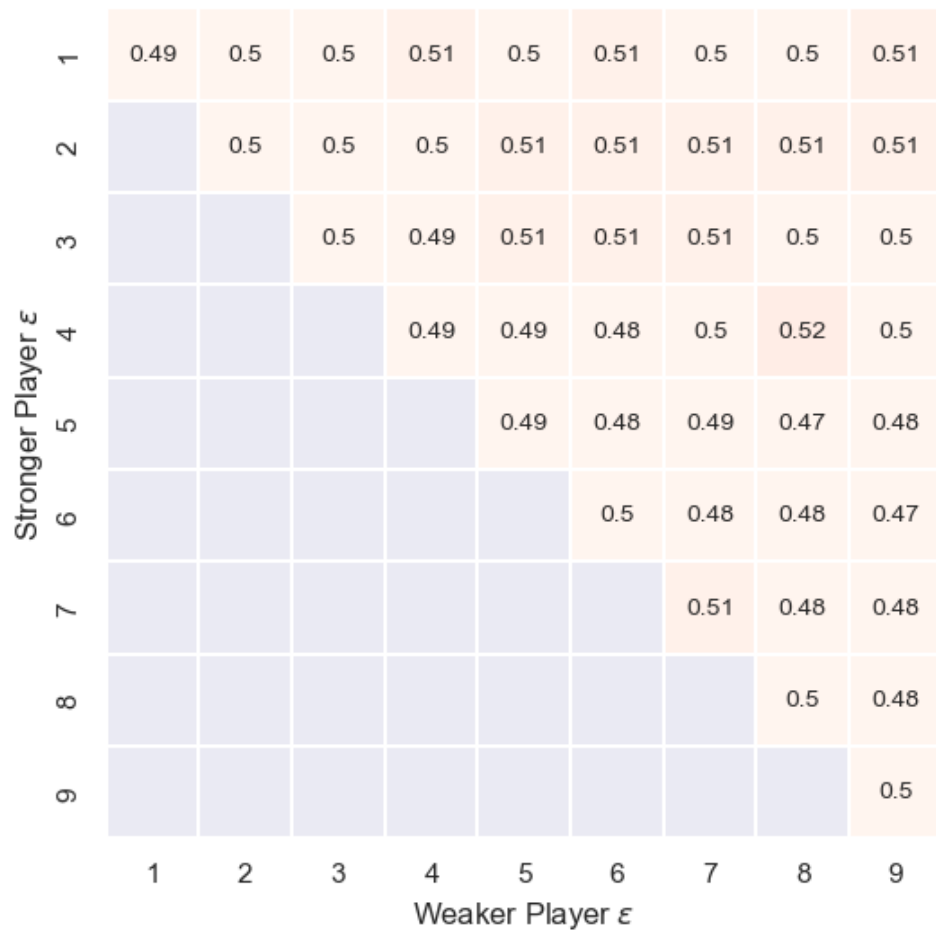
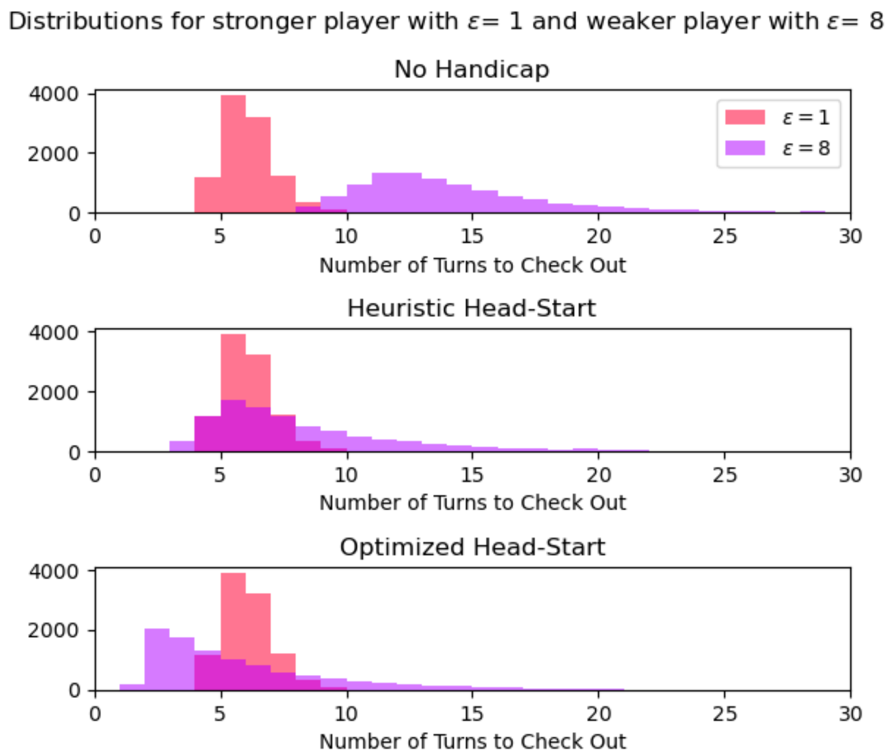
Results: Dynamic Credit Handicap



Stronger and weaker players have ~equal probability of winning

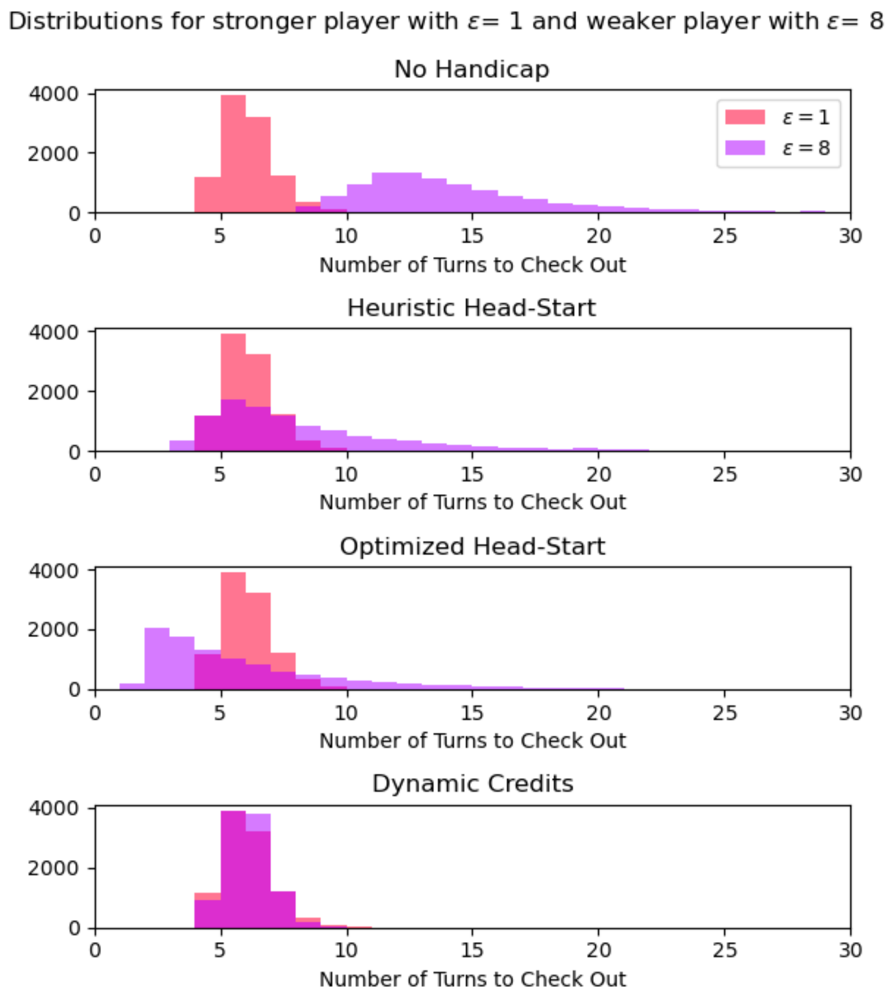
Probability that the stronger player wins after 10,000 simulated matches for different skill mismatches

Results: Dynamic Credit Handicap



Number of turns for two mismatched players in simulation with 10,000 iterations

Results: Dynamic Credit Handicap



Dynamic credits reshape the distribution to equalize win probability

Number of turns for two mismatched players in simulation with 10,000 iterations

Revisiting Competitive Balance

Competitive balance occurs when two players have an equal expected number of throws to check out

- One-player model
- Handicap by aligning expectations
- ~ 2 million states
- Single MDP
- Takes < 1 minute to solve, regular RAM sufficient

Fairness occurs when two players have an equal probability of winning

- Two-player model
- Handicap by aligning distributions
- ~ 2 <u>billion</u> states
- System of MDPs
- Takes 17 hours to solve, high RAM required

Conclusion

- Existing head-start handicap approaches are not fair
  - Main imbalance occurs at the end of the game and cannot be mitigated by a head-start
  - Heuristic head-start leaves the **weaker** player at a disadvantage
  - Optimized head-start leaves the **stronger** player at a disadvantage
  - Finding the “fair” head-start is possible but computationally impractical
  - Still possible that no optimal head-start exists

- A novel dynamic credit handicap system is a promising alternative
  - Guaranteed to always have a value that can balance competition
  - Alters the distribution of turns to check-out of weaker player so that it mimics that of the stronger player
  - Can be accurately estimated with simplified modelling framework

Thanks for listening!

Questions?

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